

Cheng Guo

CONTACT INFORMATION

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RESEARCH INTERESTS

My research develops theoretically-grounded and computationally-scalable methods to improve the operations of large-scale markets complicated by features such as nonconvexity, stochasticity, network effects, and strategic behavior, with electricity markets as a primary motivation. On the theoretical side, I study market design with provable performance guarantees and rigorous analysis, grounded in duality theory and state-of-the-art conic programming methods. On the computational side, I develop novel decomposition and convex relaxation methods for mixed-integer nonlinear, stochastic, and robust optimization, enabling the solution of large-scale market operations problems that were previously intractable.

EXPERIENCE

Clemson University, Clemson, SC December 2021 - present
School of Mathematical and Statistical Sciences
Assistant Professor, area: Operations Research (tenure clock start: Fall 2022)

Columbia University, New York, NY 2021 - 2022
Department of Industrial Engineering and Operations Research
Visiting Researcher

- Host: Daniel Bienstock
- DOE ARPA-E PERFORM (Performance-based Energy Resource Feedback, Optimization, and Risk Management) project

EDUCATION

University of Toronto, Toronto, ON September 2017 - November 2021
Department of Mechanical and Industrial Engineering
Ph.D. in Industrial Engineering, GPA: 3.96/4.00

- Advisor: Merve Bodur
- Selected coursework: Stochastic Programming & Robust Optimization, Modeling Interactions on Networks, OM Matching Markets, Mathematical Methods in Power Systems

Columbia University, New York, NY 2015 - 2017
Department of Industrial Engineering and Operations Research
M.S. in Operations Research

- Selected coursework: Transportation Analytics & Logistics, Optimization I, Programming for Financial Engineering, Seminar on Queueing Theory

Wuhan University, Wuhan, China 2011 - 2015
School of Economics and Management
B.A. in Economics, B.S. in Mathematics

- Hongyi Honor Program Outstanding Graduates Award
- Selected coursework: Advanced Microeconomics, Industrial Organization, Advanced Financial Theory, Advanced Econometrics, Chaotic Dynamical Systems, Topology

JOURNAL ARTICLES	(*: corresponding author; Underline: student coauthor) J5. <i>Copositive Duality for Discrete Energy Markets</i> [pdf] Cheng Guo*, Merve Bodur, Joshua A. Taylor Management Science 72.3 (2026): 2022-2040 J4. <i>Tightening Quadratic Convex Relaxations for the Alternating Current Optimal Transmission Switching Problem</i> [pdf] Cheng Guo*, Harsha Nagarajan, Merve Bodur INFORMS Journal on Computing, forthcoming J3. <i>Risk-Aware Security-Constrained Unit Commitment</i> [pdf] Daniel Bienstock, Yury Dvorkin, Cheng Guo*, Robert Mieth, <u>Jiayi Wang</u> IEEE Transactions on Energy Markets, Policy and Regulation 2.4 (2024): 536-551 J2. <i>Generation Expansion Planning with Revenue Adequacy Constraints</i> [pdf] Cheng Guo*, Merve Bodur, Dimitri J. Papageorgiou Computers & Operations Research 142 (2022): 105736 J1. <i>Logic-based Benders Decomposition and Binary Decision Diagram Based Approaches for Stochastic Distributed Operating Room Scheduling</i> [pdf] Cheng Guo*, Merve Bodur, Dionne M. Aleman, and David R. Urbach INFORMS Journal on Computing 33.4 (2021): 1551-1569
PREPRINTS	P3. <i>Networked Markets with Production and Edge Capacity Constraints: From Competitive Equilibria To Market Entry</i> [pdf] Cheng Guo, <u>Jiayi Wang</u>, Ozan Candogan – Conference Version: C1 (EC’26) P2. <i>Pricing Discrete and Nonlinear Markets With Semidefinite Relaxations</i> [pdf] Cheng Guo, <u>Lauren Henderson</u>, Ryan Cory-Wright, Boshi Yang P1. <i>Incentivizing Investment and Reliability: A Study on Electricity Capacity Markets</i> [pdf] Cheng Guo, Christian Kroer, Yury Dvorkin, Daniel Bienstock
WORKING PAPERS	W1. <i>Stochastic Dual Dynamic Programming for Multiobjective Multistage Problems</i> <u>Benjamin J. Hamlin</u>, Cheng Guo, Margaret Wiecek
CONFERENCE PROCEEDINGS	C1. <i>Networked Markets with Production and Edge Capacity Constraints: From Competitive Equilibria To Market Entry</i> Cheng Guo*, <u>Jiayi Wang</u>, Ozan Candogan ACM Conference on Economics and Computation (EC) (2026)
HONORS AND AWARDS	• Finalist for student Anna Deza, INFORMS Undergraduate OR Prize Competition, 2020 • Mixed Integer Programming Workshop Student Travel Support, 2019 • Hongyi Outstanding Graduates Award, 2015 • Economics and Management School Scholarship, 2013 - 2014

FUNDING	• Clemson University - Clemson Faculty SUCCEEDS: Program 1 (Project Initiation/SEED Funding, awarded to 16 PIs university-wide) (PI), 2024 • University of Toronto - Bert Wasmund Graduate Fellowships in Sustainable Energy Research, 2018
TEACHING	Clemson University <i>Instructor</i> • MATH 8100 - Mathematical Programming (graduate): Fall 2022, Spring 2023, Spring 2024, Fall 2024 • MATH 4400/6400 - Linear Programming (undergraduate/graduate): Fall 2024, Fall 2025, Spring 2026 • STAT 3090 - Introductory Business Statistics (undergraduate): Spring 2022 (virtual), Fall 2023, Fall 2024, Fall 2025 University of Toronto <i>Tutorial Teaching Assistant</i> • MIE 562 - Scheduling (undergraduate/graduate): Fall 2019, Fall 2020 • MIE 335 - Algorithms and Numerical Methods (undergraduate): Winter 2019 Wuhan University <i>Teaching Assistant</i> • Probability Theory (undergraduate): Fall 2014
ADVISING	Ph.D. Students Lauren Henderson (since 2023), Benjamin Hamlin (since 2022, co-advised with Margaret Wiecek) M.S. Students Lauren Henderson (2024) Undergraduate Students Jiayi Wang (co-advised, Columbia B.S. 2022 → Stanford Ph.D.), Anna Deza (co-advised, U. Toronto B.A.Sc. 2020 → UC Berkeley Ph.D.), Ryan Do (co-advised, U. Toronto B.A.Sc. 2019 → U. Toronto M.Eng.) Ph.D. Thesis Committee Member Kristen Joyce, Sarah Kelly (2024) M.S. Thesis Committee Member Yunheng Jiang (2022)
INVITED TALKS	• Cornell University, FIND Seminar, Ithaca, NY April, 2024 • Mixed Integer Programming Workshop, Los Angeles, CA May, 2023 • Polytechnique Montreal, GERAD Seminar, Virtual May, 2022 • Discrete Optimization Talks, Virtual December, 2020

CONFERENCE PRESENTATIONS	• (Upcoming) INFORMS MSOM Conference, Boston, MA	July, 2026
	• (Upcoming) Marketplace Innovation Workshop (MIW), Online	May, 2026
	• Production and Operations Management Society (POMS) Conference, Reno, NV	May, 2026
	• INFORMS Optimization Society Conference, Atlanta, GA	March, 2026
	• INFORMS Annual Meeting, Atlanta, GA	October, 2025
	• IEEE Power and Energy Society (PES) General Meeting, Austin, TX	July, 2025
	• International Conference on Continuous Optimization (ICCOPT), Los Angeles, CA	July, 2025
	• Production and Operations Management Society (POMS) Conference, Atlanta, GA	May, 2025
	• INFORMS Computing Society Conference, Toronto, ON	March, 2025
	• INFORMS Annual Meeting, Seattle, WA	October, 2024
	• International Symposium on Mathematical Programming, Montreal, QC	July, 2024
	• INFORMS Optimization Society Conference, Houston, TX	March, 2024
	• INFORMS Annual Meeting, Phoenix, AZ	October, 2023
	• INFORMS MSOM Conference, Montreal, QC	June, 2023
	• INFORMS Annual Meeting, Indianapolis, IN	October 2022
	• International Conference on Continuous Optimization (ICCOPT), Bethlehem, PA	July, 2022
	• INFORMS Optimization Society Conference, Greenville, SC	March, 2022
	• INFORMS Annual Meeting, Virtual	October 2021
	• International Conference on Game Theory (poster), Virtual	July 2021
	• IPCO Conference (poster), Virtual	June 2021
	• CORS Annual Conference, Virtual	June 2021
	• Mixed Integer Programming Workshop (poster), Virtual	May 2021
	• Grid Science Winter School (poster), Virtual	January 2021
	• INFORMS Annual Meeting, Virtual	November 2020
	• INFORMS Annual Meeting, Seattle, WA	October 2019
	• DIMACS Workshop on MINLP (poster), Montreal, QC	October 2019
	• Mixed Integer Programming Workshop (poster), Boston, MA	July 2019
	• Optimization Days, Montreal, QC	May 2019
	• INFORMS Computing Society Conference, Knoxville, TN	January 2019
ACADEMIC SERVICE	• Journal reviewer for <i>Operations Research</i> , <i>Mathematical Programming</i> , <i>Management Science</i> , <i>Manufacturing & Service Operations Management</i> , <i>SIAM Journal on Optimization</i> , <i>INFORMS Journal on Computing</i> , <i>Transportation Science</i> , <i>Production and Operations Management</i> , <i>Computers and Operations Research</i> , <i>IEEE Transactions on Power Systems</i>	
	• Conference reviewer for <i>EC (2026)</i> , <i>IEEE Power & Energy Society General Meeting (2024, 2025, 2026)</i>	
	• Vice President/President-Elect for INFORMS Junior Faculty Interest Group (JFIG) (2025-2027)	
	• Program committee member for Mixed Integer Programming (MIP) Workshop 2026	

	• Session Chair for INFORMS Optimization Society Conference (2022, 2024, 2026); INFORMS Annual Meeting (2019, 2021, 2022, 2023, 2024); CORS Annual Meeting (2021)	
UNIVERSITY SERVICE	• Clemson University SMSS Research Committee (2023-present) • Co-organizer of University of Toronto MIE UTOrg Seminar (2019-2020)	
ADVANCED TRAINING	• “Cultivating an Inclusive Classroom Environment”, Academic Impressions	2024
SKILLS	• Programming language: Python, Julia, C++ • Software: Gurobi, CPLEX, Knitro, Mosek	
OTHER ACTIVITIES	• INFORMS UofT Student Chapter (Honorable Mention, 2020), Vice President (2019-2021) • Columbia IEOR Mentorship Program, Mentor (2018-2020) • Wuhan U. Women Soccer Team, Captain (2013-2015)	